Exception Handling Rules for FreeST

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1 Syntax

Multiplicity	$m ::= \text{un} \mid \text{lin}$
Basic Kinds	$\upsilon ::= \mathbf{C} \mid \mathbf{S} \mid \mathbf{T}$
Kinds	$\kappa ::= \upsilon^m$
Polarity	$\sharp ::= m! \mid m?$
View	$\star ::= m\oplus \mid m\&$
Channel End	$\text{End} ::= \text{Close} \mid \text{Wait}$
Types	$T, U ::= \text{Skip} \mid \sharp T \mid \star\{\ell\colon T_\ell\}_{\ell\in \text{L}} \mid \text{End} \mid T;T$
	$()_m \mid T \times T \mid T \to_m T \mid \{\ell\colon T_\ell\}_{\ell\in \text{L}} \mid \langle \ell\colon T_\ell\rangle_{\ell\in \text{L}}$
	$\forall a\colon \kappa\ .\ T \mid \mu a\colon \kappa.T \mid a$
Kinding contexts	$\Delta ::= \varepsilon \mid \Delta, a\colon \kappa$
Variables	x, y
Terms	$L, M, N ::= x \mid \lambda_m x\colon T.M \mid M\ N \mid ()_m \mid (M, N)$
	$\textbf{let}\ ()_m = M\ \textbf{in}\ N \mid \textbf{let}\ (x, y) = M\ \textbf{in}\ N$
	$\textbf{channel}\ @T \mid \textbf{fork}\ M \mid \textbf{send}\ M\ N \mid \textbf{receive}\ M$
	$\textbf{sendType}\ @T\ M \mid \textbf{receiveType}\ M$
	$\textbf{select}\ \ell \mid \textbf{case}\ M\ \textbf{of}\ \{m\&\ell_i\ M_i \to N_i\}^{i\in 1..n}$
	$\textbf{close}\ M \mid \textbf{wait}\ M \mid \textbf{cancel}\ M \mid \textbf{raise}$
	$\textbf{try}\ L\ \textbf{as}\ x\ \textbf{in}\ M\ \textbf{otherwise}\ N$
Typing contexts	$\Gamma ::= \varepsilon \mid \Delta, x\colon T$
Configurations	$\mathcal{C}, \mathcal{D}, \mathcal{E} ::= (\nu a)\mathcal{C} \mid \mathcal{C} \parallel \mathcal{D} \mid \lightning a \mid a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}),\ p, q \geq 0$
Evaluation contexts	$\mathcal{F} ::= \ldots$
Pure contexts	$P ::= \ldots$
Configuration contexts	$\mathcal{G} ::= [\,] \mid (\nu a)\mathcal{G} \mid \mathcal{G} \parallel \mathcal{C}$

2 Typing Rules

$$\text{T-CANCEL}\ \frac{\Gamma \vdash M : T \qquad \Delta \vdash T : \mathbf{S}^{\mathsf{lin}}}{\Gamma \vdash \mathbf{cancel}\ M : ()} \qquad \text{T-RAISE}\ \frac{}{\vdash \mathbf{raise} : T} \qquad \text{T-TRYIN}\ \frac{\Gamma_1 \vdash L : T_1 \qquad \Gamma_2, x : T_1 \vdash M : T_2 \qquad \Gamma_2 \vdash N : T_2}{\Gamma_1, \Gamma_2 \vdash \mathbf{try}\ L\ \mathbf{as}\ x\ \mathbf{in}\ M\ \mathbf{otherwise}\ N : T_2}$$

3 Reduction Rules

Term Reduction $\boxed{M \to_M N}$

E-TRYINVAL $\quad \mathbf{try}\ V\ \mathbf{as}\ x\ \mathbf{in}\ M\ \mathbf{otherwise}\ N \ \to_M \ M\{V/x\}$

Configuration Equivalence $\boxed{\mathcal{C} \equiv \mathcal{D}}$

$$\mathcal{C} \parallel (\mathcal{D} \parallel \mathcal{E}) \equiv (\mathcal{C} \parallel \mathcal{D}) \parallel \mathcal{E} \qquad \mathcal{C} \parallel \mathcal{D} \equiv \mathcal{D} \parallel \mathcal{C} \qquad (va)(vb)\mathcal{C} \equiv (vb)(va)\mathcal{C}$$

$$\mathcal{C} \parallel (va)\mathcal{D} \equiv (va)(\mathcal{C} \parallel \mathcal{D}), \quad \text{if } a \notin \mathsf{fn}(\mathcal{C})$$

$$a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}) \equiv b(\overrightarrow{W})\ q \leftrightsquigarrow p\ a(\overrightarrow{V}) \qquad () \parallel \mathcal{C} \equiv \mathcal{C}$$

$$(va)(vb)(a(\varepsilon)\ 0 \leftrightsquigarrow 0\ b(\varepsilon)) \parallel \mathcal{C} \equiv \mathcal{C}$$

Configuration Reduction $\boxed{\mathcal{C} \to \mathcal{D}}$

E-CHANNEL $\quad \mathcal{F}\ [\mathbf{channel}\ @U] \to (va)(vb)(\mathcal{F}\ [(a,b)]) \parallel a(\varepsilon)\ 0 \leftrightsquigarrow 0\ b(\varepsilon), \quad$ where a, b are fresh

E-FORK $\quad \mathcal{F}\ [\mathbf{fork}\ (\lambda().M)] \to \mathcal{F}\ [()] \parallel M$

E-SEND $\quad \mathcal{F}\ [\mathbf{send}_m\ U\ a] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}) \to \mathcal{F}\ [a] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W} \cdot U)$

E-RECEIVE $\quad \mathcal{F}\ [\mathbf{receive}_m\ a] \parallel a(U \cdot \overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}) \to \mathcal{F}\ [(U,a)] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W})$

E-SENDTYPE $\quad \mathcal{F}\ [\mathbf{sendType}\ @U\ a] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}) \to \mathcal{F}\ [a] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W} \cdot @U)$

E-RECEIVETYPE $\quad \mathcal{F}\ [\mathbf{receiveType}\ a] \parallel a(@U \cdot \overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}) \to \mathcal{F}\ [(@U,a)] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W})$

E-CLOSE $\quad \mathcal{F}\ [\mathbf{close}\ a] \parallel a(\varepsilon)\ 1 \leftrightsquigarrow 1\ b(\overrightarrow{W}) \to \mathcal{F}\ [()] \parallel a(\varepsilon)\ 0 \leftrightsquigarrow 1\ b(\overrightarrow{W} \cdot \mathrm{EOS})$

E-WAIT $\quad (va)(vb)(\mathcal{F}\ [\mathbf{wait}\ a] \parallel a(\mathrm{EOS})\ 1 \leftrightsquigarrow 0\ b(\varepsilon)) \to \mathcal{F}\ [()]$

E-SELECT $\quad \mathcal{F}\ [\mathbf{select}_m\ \ell\ a] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}) \to \mathcal{F}\ [a] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W} \cdot \ell)$

E-CASE $\quad \mathcal{F}\ [\mathbf{case}\ a\ \mathbf{of}\ \{m\&\ell_i\ a_i \to M_i\}^{i \in 1..n}] \parallel a(\ell_j \cdot \overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}) \to \mathcal{F}\ [M_j] \parallel a_j(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W})$

E-CANCEL $\quad \mathcal{F}\ [\mathbf{cancel}\ a] \parallel a(\overrightarrow{V})\ p \leftrightsquigarrow q\ b(\overrightarrow{W}) \to \mathcal{F}\ [()] \parallel \lightning a \parallel a(\overrightarrow{V})\ (p-1) \leftrightsquigarrow q\ b(\overrightarrow{W})$

E-ZAP $\quad \lightning a \parallel a(U \cdot \overrightarrow{V})\ 0 \leftrightsquigarrow q\ b(\overrightarrow{W}) \to \lightning a \parallel \lightning U \parallel a(\overrightarrow{V})\ 0 \leftrightsquigarrow q\ b(\overrightarrow{W})$

E-RECEIVEZAP $\quad \mathcal{F}\ [\mathbf{receive}_m\ a] \parallel \lightning b \parallel a(\varepsilon)\ p \leftrightsquigarrow 0\ b(\overrightarrow{W}) \to \mathcal{F}\ [\mathbf{raise}] \parallel \lightning b \parallel a(\varepsilon)\ (p-1) \leftrightsquigarrow 0\ b(\overrightarrow{W})$

E-RECEIVETYPEZAP $\quad \mathcal{F}\ [\mathbf{receiveType}\ a] \parallel \lightning b \parallel a(\varepsilon)\ 1 \leftrightsquigarrow 0\ b(\overrightarrow{W}) \to \mathcal{F}\ [\mathbf{raise}] \parallel \lightning a \parallel \lightning b \parallel a(\varepsilon)\ 0 \leftrightsquigarrow 0\ b(\overrightarrow{W})$

E-WAITZAP $\quad \mathcal{F}\ [\mathbf{wait}\ a] \parallel \lightning b \parallel a(\varepsilon)\ 1 \leftrightsquigarrow 0\ b(\overrightarrow{W}) \to \mathcal{F}\ [\mathbf{raise}] \parallel \lightning a \parallel \lightning b \parallel a(\varepsilon)\ 0 \leftrightsquigarrow 0\ b(\overrightarrow{W})$

E-CASEZAP $\quad \mathcal{F}\ [\mathbf{case}\ a\ \mathbf{of}\ \{m\&\ell_i\ a_i \to M_i\}^{i \in 1..n}] \parallel \lightning b \parallel a(\varepsilon)\ p \leftrightsquigarrow 0\ b(\overrightarrow{W}) \to \mathcal{F}\ [\mathbf{raise}] \parallel \lightning b \parallel a(\varepsilon)\ (p-1) \leftrightsquigarrow 0\ b(\overrightarrow{W})$

E-TRYINRAISE $\quad \mathcal{F}\ [\mathbf{try}\ P\ [\mathbf{raise}]\ \mathbf{as}\ x\ \mathbf{in}\ M\ \mathbf{otherwise}\ N] \to \mathcal{F}\ [N] \parallel \lightning P$

E-RAISE $\quad P\ [\mathbf{raise}] \to \lightning P$

E-LIFT $\quad \mathcal{G}\ [\mathcal{C}] \to \mathcal{G}\ [\mathcal{D}], \quad$ if $\mathcal{C} \to \mathcal{D}$